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1. Introduction

Modern financial markets are characterized by immense complexity, volatility, and susceptibility to numerous macroeconomic and geopolitical factors. For years, investors, analysts, and financial institutions have been seeking increasingly precise tools to forecast future changes in stock index values, which is crucial for making sound investment decisions. In light of the growing availability of data and technological advancements, methods based on artificial intelligence and machine learning are becoming increasingly important.

The aim of this study is to examine the potential for predicting the values of selected stock market indices using machine learning models, with a particular focus on LSTM (Long Short-Term Memory) recurrent neural networks. The analysis utilized historical data, covering not only the index values themselves but also key economic indicators, such as oil and gold prices. The use of this type of data was intended to enhance the model's effectiveness by accounting for a broader market context.

The thesis includes both a theoretical section, discussing the fundamentals of sequential models and methods for training them, and a practical section—covering data preparation, the construction of a predictive model, and the evaluation of its effectiveness. The end result is a tool capable of forecasting short-term changes in stock market indices, which can serve as a starting point for further research and practical applications in the capital market.

Additionally, as part of the exploratory analysis, we visualized the correlations between stock market indices (such as the S&P 500, FTSE 100, DAX, HSI, AORD, and N225) and their interdependencies to better understand the structure of the model's input data.

2. Data Sources

The data used in the analysis comes from publicly available online sources. The goal was to collect both market data (stock index prices) and macroeconomic indicators affecting financial markets. Detailed list of sources:

- Gold price - https://github.com/datasets/gold-prices/tree/main
- Oil price - https://fred.stlouisfed.org/series/DCOILWTICO
- FED interest rate - https://fred.stlouisfed.org/series/FEDFUNDS
- US inflation - https://fred.stlouisfed.org/series/FPCPITOTLZGUSA
- Real US GDP - https://fred.stlouisfed.org/series/GDPC1
- US unemployment rate - https://fred.stlouisfed.org/series/UNRATE

Stock index data was retrieved using the Yahoo Finance API (https://finance.yahoo.com) with the yfinance library in Python. The list of analyzed indices includes:

- S&P 500 (US) - An index comprising the 500 largest companies listed on the NYSE and NASDAQ, considered one of the best indicators of the health of the US stock market.
- FTSE 100 (United Kingdom) – An index of the 100 largest companies listed on the London Stock Exchange by market capitalization.
- DAX (Germany) – An index of the 40 largest German companies listed on the Frankfurt Stock Exchange.
- HSI (Hang Seng Index) (Hong Kong) – The main stock market index in Hong Kong, reflecting the performance of the largest companies listed on that market.
- AORD (All Ordinaries) (Australia) - One of the main indices of the Australian Stock Exchange (ASX), comprising approximately 500 of the largest companies.
- N225 (Nikkei 225) (Japan) - The most important index of the Tokyo Stock Exchange, tracking the share prices of 225 selected companies.

3. Data Preparation

Each index was filtered so that only the column showing the closing price for each trading day remained. The data was then sorted by date.

Data on macroeconomic indicators (such as inflation, gold prices, oil prices, interest rates, GDP, and the unemployment rate) were downloaded in CSV format from the FRED (Federal Reserve Economic Data) portal and from public repositories (in the case of gold prices). Next, using the pandas library, this data was loaded into the analytical environment, standardized, and combined into a single dataset along with the stock index data.

The data was prepared to cover a common time period: from January 4, 1988, to December 31, 2024, which ensured the consistency of the analysis and enabled the model to be trained on historical data using a range of potentially significant variables affecting stock index prices.

Table 1. Final data

	Fed rate	Oil price	Inflation	GDP	Gold price	Unemployment	S&P 500	FTSE 100	DAX	HSI	AORD	N225
Date												
01/04/1988	7.3	17.77	4.37	9,367.5	476.58	5.7	255.94	1,713.9	956.49	2,286.3	1,297.4	21,217.04
01/05/1988	7.34	17.89	4.37	9,367.5	476.58	5.7	258.63	1,789.6	996.1	2,403.2	1,309.3	21,575.28
01/06/1988	6.94	17.73	4.37	9,367.5	476.58	5.7	258.89	1,787.1	1006.01	2,463	1,321.9	22,790.5
01/07/1988	6.91	17.3	4.37	9,367.5	476.58	5.7	261.07	1,787.2	1,014.47	2,452.5	1,316.6	22,792.13
01/08/1988	6.8	17.33	4.37	9,367.5	476.58	5.7	243.4	1,773.4	1026.69	2,460.7	1,309.5	22,872.56
...	...	...	...	...	...	...	...	...	...	...	...	...
12/24/2024	4.33	70.87	3.75	23,536.29	2,648.01	4.1	6,040.04	8,137	19,848.77	20,098.29	8,471.5	39,036.85
12/26/2024	4.33	70.38	3.75	23,536.29	2,648.01	4.1	6,037.59	8,137	19,848.77	20,098.29	8,471.5	39,568.06
12/27/2024	4.33	71.28	3.75	23,536.29	2,648.01	4.1	5,970.84	8,149.8	19,984.32	20,090.46	8,520.1	40,281.16
12/30/2024	4.33	71.73	3.75	23,536.29	2,648.01	4.1	5,906.94	8,121	19,909.14	20,041.42	8,496	39,894.54
12/31/2024	4.33	72.44	3.75	23,536.29	2,648.01	4.1	5,881.63	8,173	19,909.14	20,059.95	8,420.5	39,894.54

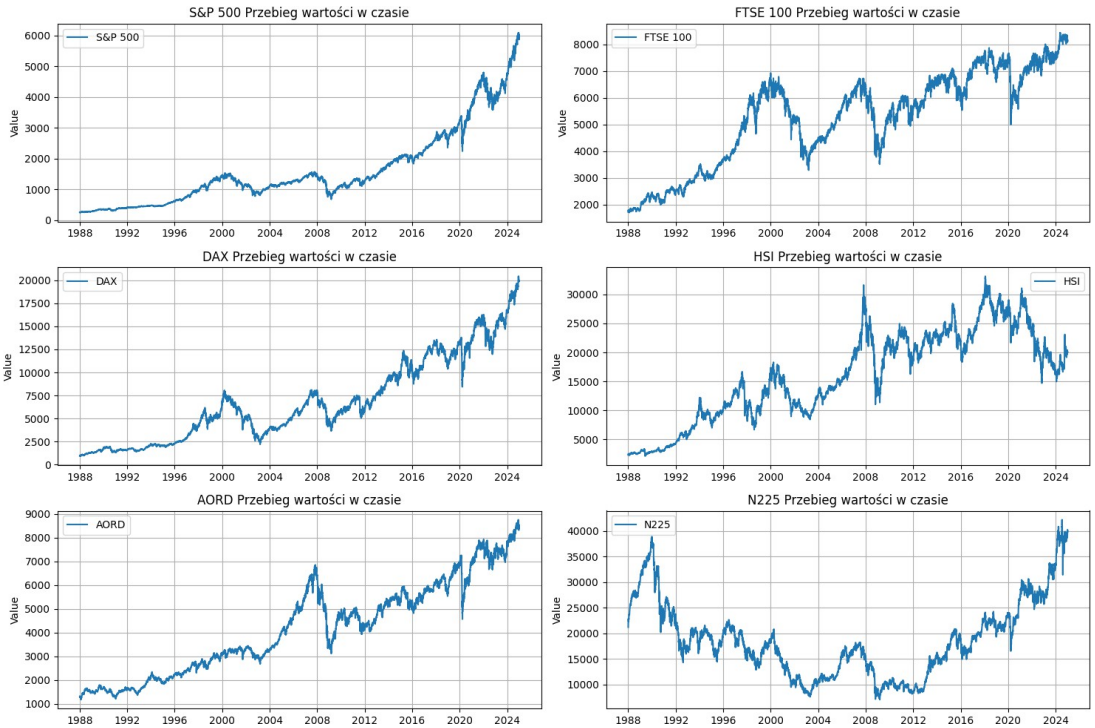
Source: own analysis

4. Preliminary data analysis and visualization

4.1 Analysis of stock market index trends

The first step in the analysis was to present the trends of selected stock market indices over the past decades, as shown in the charts below.

Figure 1. Stock market indices—trends over time



Source: matplotlib, own work

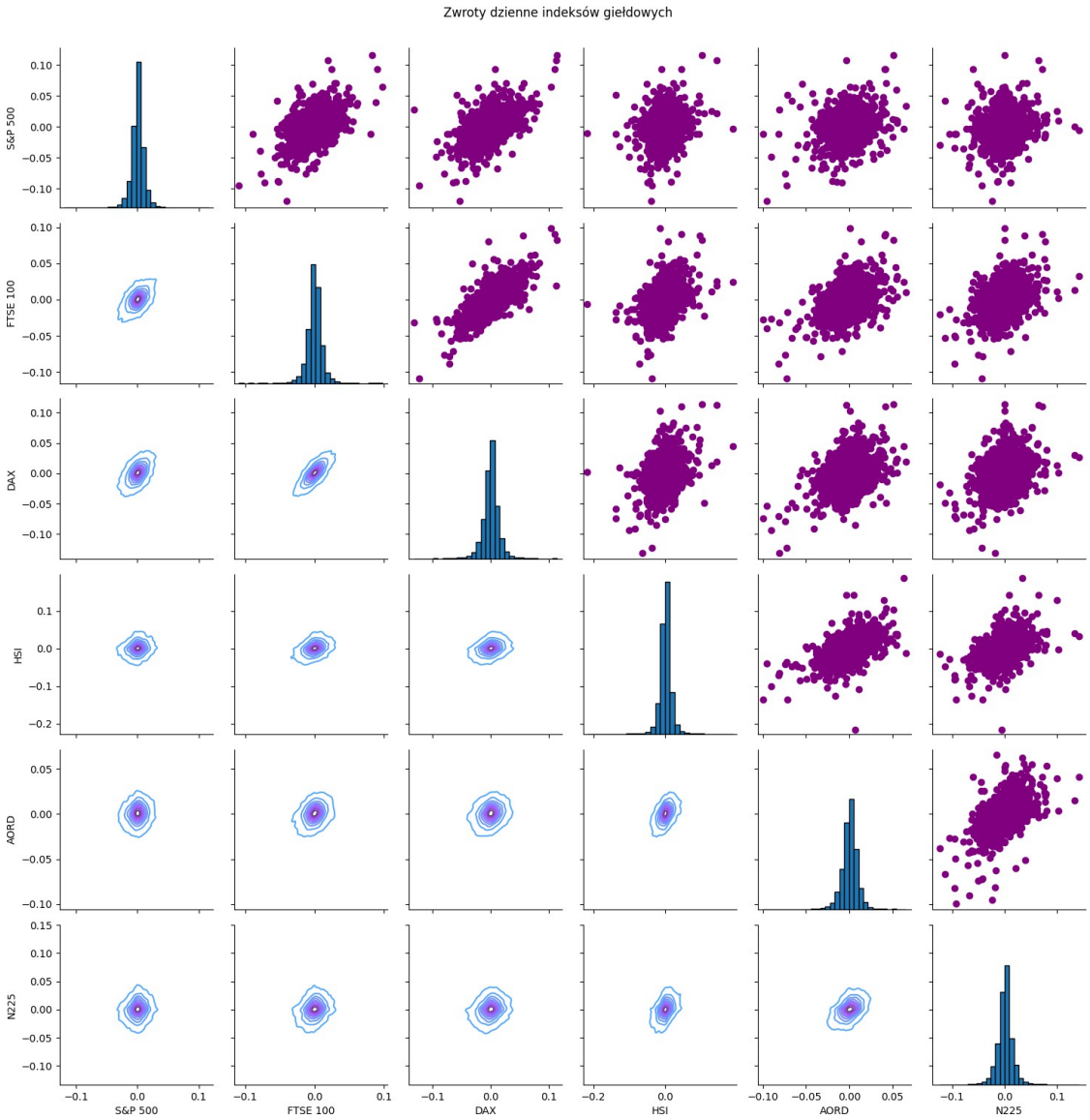
The chart shows that most indices—with the exception of the N225—exhibit a strong upward trend. This is the result of long-term economic growth. The more the economy grows, the more stock market indices rise, reflecting the health of the capital market.

The N225 index (Japanese stock market) provides a good reflection of the country's economic situation during the period under review. Following dynamic economic growth after World War II, Japan began to grapple with prolonged deflation in the 1990s, which resulted in an economic slowdown. As a result, the Japanese stock market remained well below its historical highs for many years.

4.2 Correlation of Daily Index Returns

To better understand the relationships between the indices, average daily returns were analyzed. The correlations between the returns of individual indices are presented in the form of scatter plots and histograms.

Figure 2. Daily returns of stock market indices



Source: matplotlib, seaborn, own work

Along the diagonal of the matrix are histograms of the distributions of daily returns for each index. All distributions are similar in shape to a normal distribution, with clearly defined "tails," indicating the presence of extreme events (large declines or increases), typical of financial data.

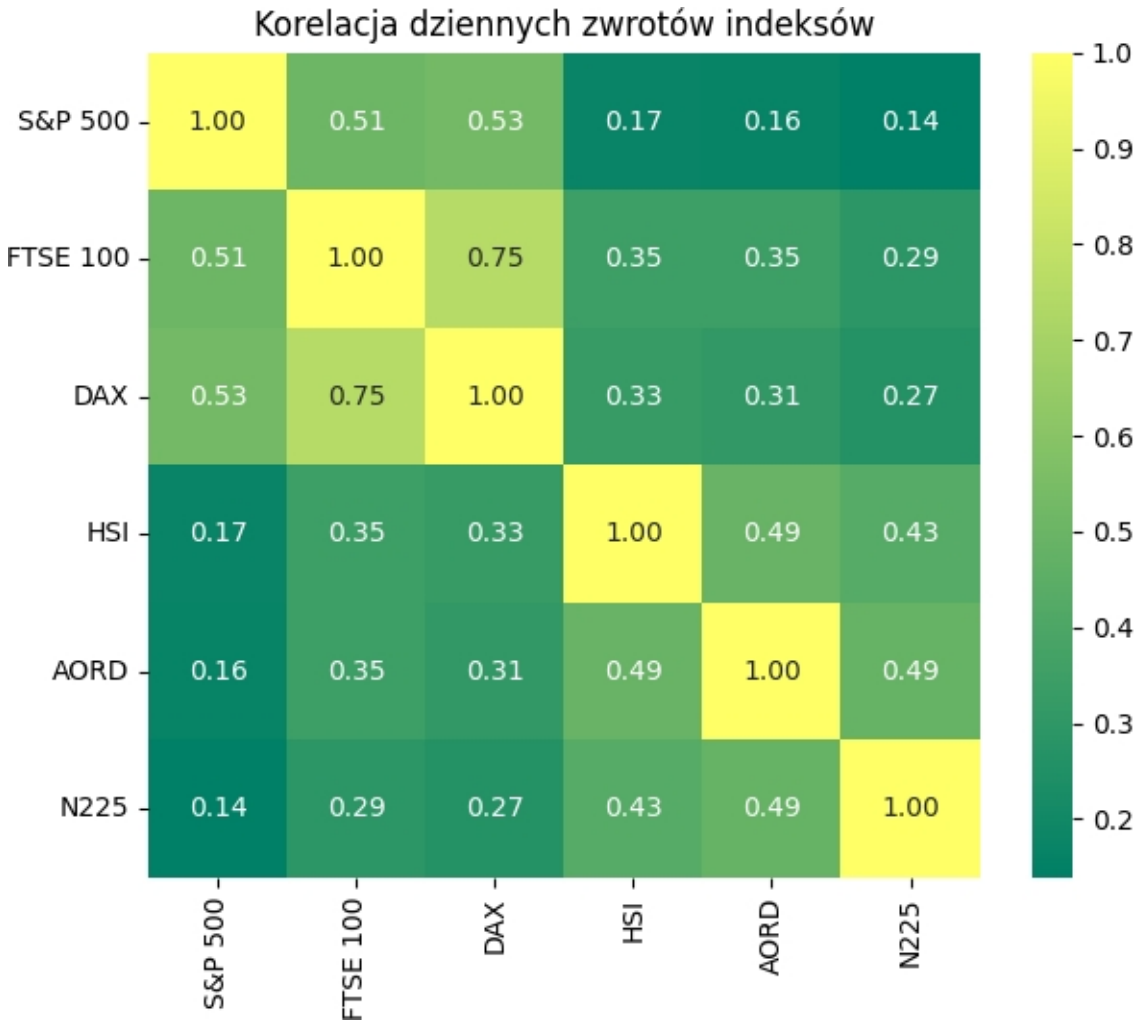
The remaining fields of the matrix present scatter plots illustrating the interdependencies between pairs of indices. In many cases, oblique, elliptical clusters of points can be observed, indicating positive correlations between the daily returns of individual markets. Particularly strong correlations exist between the FTSE 100 and DAX (European markets).

Such relationships are natural and result from the global integration of financial markets—stock exchanges react to the same macroeconomic information, geopolitical data, and signals from the world's largest economies. For less correlated index pairs, such as the N225 and S&P 500, the charts show more dispersed clusters, indicating a weaker relationship between these markets.

Some of the plots have been supplemented with density contours to facilitate the identification of areas of highest data concentration. These contours confirm that most returns are concentrated around values close to zero, which is consistent with the typical characteristics of financial markets.

To better illustrate the relationships between the indices, a correlation matrix of daily returns was also used, which is presented in Figure 3.

Figure 3. Correlation matrix of daily stock index returns



Source: matplotlib, seaborn, own work

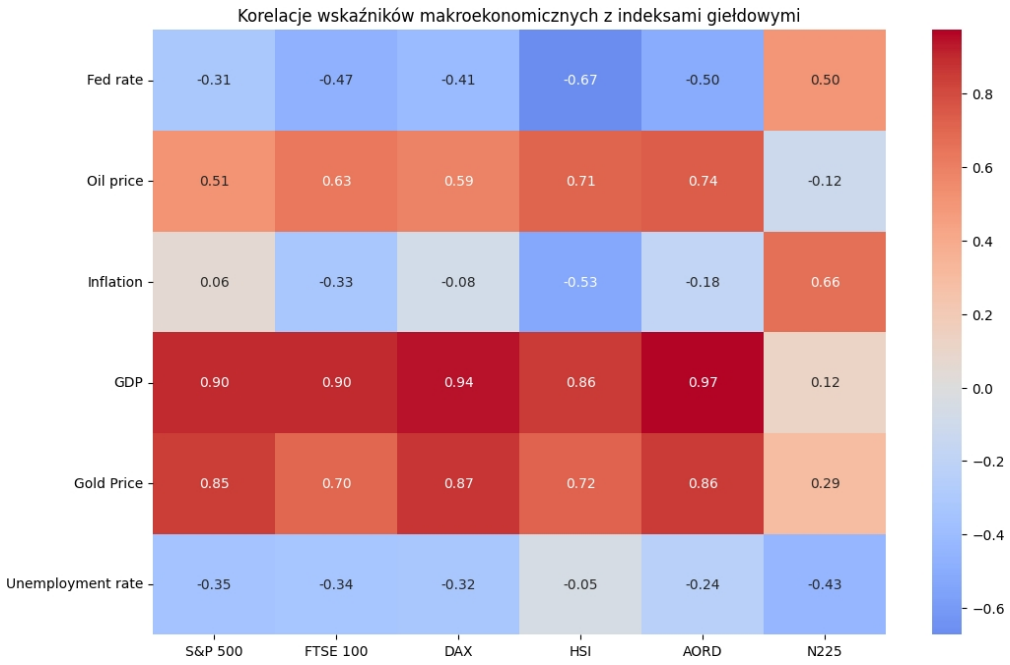
4.3 Correlations between macroeconomic indicators and indices

The subsequent part of the analysis examined the relationships between stock market indices and selected macroeconomic indicators (such as inflation, unemployment, and the price of gold). These indicators primarily concern the United States, as the U.S.—being the world's largest economy—significantly influences global markets and sets the pace for other countries. For this reason, detailed data on other countries have been omitted.

The figure below shows a correlation matrix between macroeconomic indicators and stock market indices. The correlations were calculated using Pearson's correlation coefficient, with values ranging

-1 to 1. Positive values indicate a direct relationship, while negative values indicate an inverse relationship.

Figure 4. Correlation matrix of macroeconomic indicators with stock market indices



Source: matplotlib, seaborn, own work

The highest positive correlations were observed between GDP and stock market indices, confirming that the stock market grows alongside economic growth. In the case of Japan, no correlation exists, which can be explained by the deflation described earlier.

A high positive correlation was also observed with respect to the price of gold. This is somewhat paradoxical, as gold is considered a "safe haven," and its price typically rises during periods of economic uncertainty. In practice, however, the price of gold can also rise during economic booms, which is a result of inflation and growing demand for precious metals as a form of capital preservation.

The price of oil also shows a significant positive correlation—particularly with the AORD (0.74), HSI (0.71), and DAX (0.59) indices. This stems from the sensitivity of these economies to changes in energy commodity prices.

In contrast, the strongest negative correlations were observed with the Fed's interest rate, particularly relative to the HSI (-0.67), AORD (-0.50), and FTSE 100 (-0.47) indices. This indicates that interest rate hikes in the U.S. may negatively impact global capital markets, among other things by limiting the availability of cheap capital. The unemployment rate also shows a negative correlation, which is in line with expectations—rising unemployment is typically associated with a slowdown in economic activity and stock market declines.

The impact of inflation is more varied—in some cases, it shows a negative correlation (e.g., HSI: -0.53), and in others, a positive one (e.g., N225: 0.66). This may stem from differences in monetary policy and the purchasing power of individual countries' currencies.

5. Building the predictive model

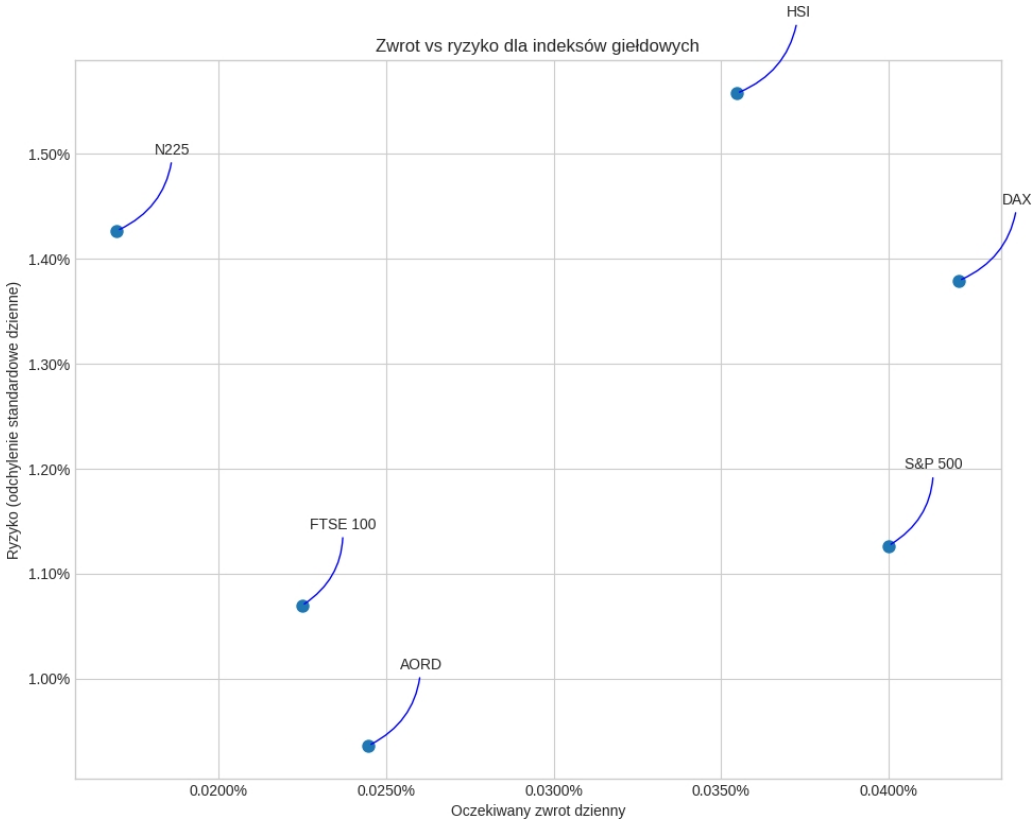
5.1. Selection of indices for prediction and choice of model

For the purposes of this analysis, we decided to use data from six major stock indices representing different regions of the world:

- S&P 500 (USA),
- DAX (Germany),
- FTSE 100 (United Kingdom),
- HSI (Hong Kong),
- NIKKEI 225 (Japan),
- AORD (Australia).

The objective was to identify the indices that demonstrate the greatest stability and investment attractiveness—in terms of daily returns and risk. Based on statistical analysis, a graph was created showing the relationship between expected daily returns and standard deviation (risk), which allowed for a comparison of the characteristics of the analyzed indices.

Figure 5. Return vs. Risk for Stock Market Indices



Source: matplotlib, own work

Based on the chart above, it can be observed that:

- The S&P 500 is characterized by the highest expected daily return with a relatively low level of risk,
- The DAX also exhibits a high level of expected return, with a slightly higher but still predictable level of volatility,
- The remaining indices, despite certain advantages, were characterized by lower returns or higher risk, which limited their usefulness in the context of effective prediction using sequential models.

As a result, the S&P 500 and the DAX were selected as the primary targets for prediction.

5.2. Description of the LSTM Model Used

A model based on a Long Short-Term Memory (LSTM) recurrent neural network was used to predict stock index values. LSTM is a variant of Recurrent Neural Networks (RNNs) designed for the effective processing of time series. Unlike classical RNNs, the LSTM architecture eliminates the vanishing gradient problem through the use of special mechanisms: forget, input, and output gates. This allows the model to "remember" long-term dependencies in time-series data, which is particularly important for forecasting financial markets.

The input data for the model was prepared in the form of 60-day time windows, based on which the model learned to predict the index value for the following day. The model used two LSTM layers and a Dense layer as the output layer.

The model initialization code (in simplified form) looked as follows:

Figure 6. Model initialization code

```
model = Sequential()
model.add(LSTM(128, return_sequences=True, input_shape=(x_train.shape[1], x_train.shape[2])))
model.add(LSTM(64, return_sequences=False))
model.add(Dense(25))
model.add(Dense(1))

model.compile(optimizer='adam', loss='mean_squared_error')
model.fit(x_train, y_train, batch_size=32, epochs=20)
```

Source: own work

Layer description:

- First LSTM layer (128 units) – analyzes data sequences while retaining information about long-term dependencies. It returns the entire sequence to enable subsequent layers to analyze the temporal context.
- Second LSTM layer (64 units) – acts as a filter for relevant information from previous layers, returning only the latest state, which represents the input sequence.
- Dense layer (25 neurons) – serves as a transitional layer, preparing the data for the final prediction.

- Final Dense layer (1 neuron) – returns a single output value: the predicted value of the S&P 500 index.
- The batch size—the number of samples processed before updating the model weights—and the number of epochs were tested in various configurations. The best results were obtained for: batch size = 32 and number of epochs = 20.

The training process was conducted on a dataset covering the period from 1988-01-04 to 2023-02-24. The data was normalized using the MinMaxScaler, which improved the model's stability. The mean squared error (MSE) was adopted as the cost function, and the Adam algorithm was used for optimization.

5.3. Selection of macroeconomic indicators

Initially, the model was trained using a full set of macroeconomic variables, including:

- gold prices
- oil prices
- inflation
- interest rates
- GDP
- unemployment

However, upon further analysis—including an examination of the correlations between these indicators and stock market indices—it was observed that many of these variables introduced noise and degraded the model's predictive power. For example, GDP, despite exhibiting a very high correlation, did not add value to the model.

The analysis showed that the price of gold and the price of crude oil (excluding GDP) had the highest positive correlations with stock market indices and had a positive impact on the learning process. After reducing the input set to only these two variables, the model not only learned faster but also achieved significantly better predictive results

(lower error on the test set). Furthermore, the absence of redundant data reduced the risk of overfitting.

6. Model evaluation and analysis of prediction results

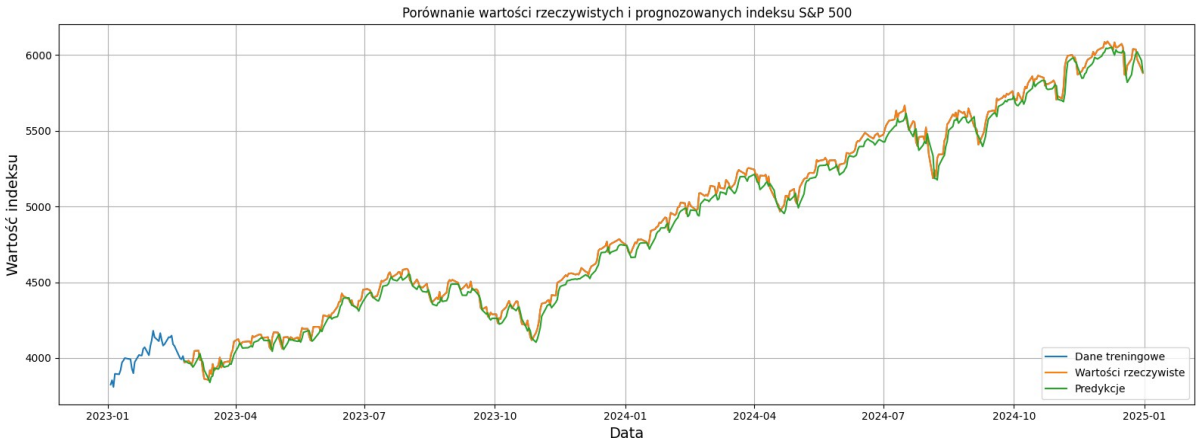
6.1. Evaluation of the LSTM model's effectiveness

To evaluate the effectiveness of the proposed LSTM model, predictions were made for the S&P 500 and DAX stock indices for the period from January 2023 to the end of December 2024. The predictions were compared with the actual index values, and the results are presented in the charts. To better illustrate the model's performance, the charts also include a portion of the training data used to train the model.

The following symbols are used in the figures:

- blue - actual index values in the training set,
- orange - actual index values in the test set,
- green – values predicted by the model.

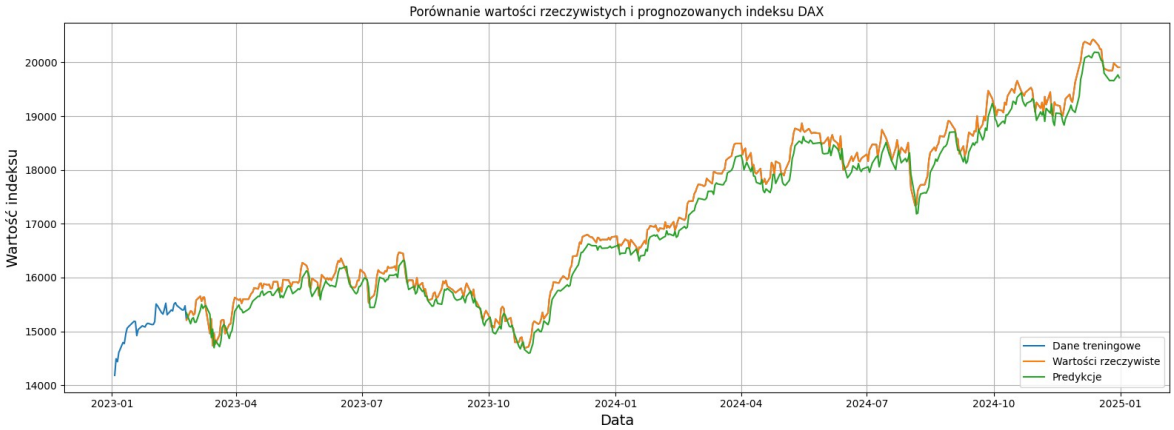
Figure 7. Comparison of actual and predicted values of the S&P 500 index



Source: matplotlib, seaborn, own work

The model very accurately captured the trend of the actual S&P 500 index values. There is a significant convergence between the prediction curve and the actual data, which confirms the model's high effectiveness in identifying and tracking major market trends.

Figure 8. Comparison of actual and forecast values of the DAX index



Source: matplotlib, seaborn, own work

High prediction accuracy was also achieved for the DAX index. The model effectively captured not only the direction of changes but also the dynamics of the index's fluctuations. The forecasts accurately reflect both upward and downward trends, confirming the model's robustness even under volatile market conditions.

6.2. Comparison of Rates of Return

To further assess the model's quality, the rates of return from actual and forecasted indices were analyzed during the test period, February 24, 2023 – December 31, 2024.

Table 2. Rate of return for the S&P 500

S&P 500 Real Rate of Return	S&P 500 predicted rate of return
67.50%	67.61%

Table 3. Rate of return for the DAX

DAX real rate of return	DAX predicted rate of return
76.40%	77.84%

The results of the analysis indicate that the differences between the predicted and actual values fall within the margin of statistical error, which means that the model is highly likely to be accurate.

7. Conclusions and summary

The experiments and analyses conducted confirmed that sequential models such as LSTM can be successfully applied to short-term forecasting of stock market indices such as the DAX or S&P 500.

The results obtained showed that the model accurately captures changes in index values under relatively stable and predictable market conditions. This is evidenced by the very good agreement between the forecasted values and the actual data during the test period, as presented in the relevant charts in Chapter 6.

It should be noted, however, that the model's effectiveness may drop significantly in exceptional situations, such as global crises, sudden political events, or unpredictable economic phenomena. Examples include the COVID-19 pandemic, the outbreak of war, or the current geopolitical realities, in which a single post on a social media platform or a political decision—such as Donald Trump's announcement of tariffs—can trigger rapid changes in financial markets within a short period of time. Such phenomena are difficult to capture using models based on historical data, as they are neither seasonal nor repetitive in nature.

Despite these limitations, the application of the LSTM model in stock market data analysis has proven effective and can serve as a valuable tool to support investment decisions, especially under conditions of relative market stability. The final investment decision should, however, also take into account other factors, including fundamental analysis and current events.

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